

Rational Expressions & Equations

DSAT Math Practice • 20 Questions • Mixed Difficulty

Total Questions: 20 | Multiple Choice: 16 | Grid-In (SPR): 4

Question 1 [Easy]

Which expression is equivalent to $(x^2 - 9) / (x + 3)$?

- A) $x - 3$
 - B) $x + 3$
 - C) $x^2 - 3$
 - D) $(x - 3)(x + 3)$
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Question 2 [Easy]

Simplify: $(6x^3) / (2x)$.

- A) $3x^2$
 - B) $4x^2$
 - C) $3x^3$
 - D) $6x^2$
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Question 3 [Easy]

What value of x makes the expression $(x + 5) / (x - 2)$ undefined?

- A) -5
 - B) 2
 - C) -2
 - D) 5
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Question 4 [Easy]

Which expression is equivalent to $3/(x + 1) + 2/(x + 1)$?

- A) $5/(x + 1)$
 - B) $5/(2x + 2)$
 - C) $6/(x + 1)^2$
 - D) $5/(x + 1)^2$
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Question 5 [Medium]

Simplify: $(x^2 - 4x - 12) / (x^2 - 36)$.

- A) $(x + 2)/(x + 6)$
 - B) $(x - 2)/(x - 6)$
 - C) $(x + 2)/(x - 6)$
 - D) $(x - 6)/(x + 2)$
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Question 6 [Easy]

Student-Produced Response

If $5/(x - 3) = 10$, what is the value of x ?

Question 7 [Medium]

Which of the following is equivalent to $(2x + 6)/(x^2 + 5x + 6)$?

- A) $2/(x + 2)$
 - B) $2/(x + 3)$
 - C) $(2x + 6)/(x + 2)$
 - D) $2x/(x + 2)$
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Question 8 [Medium]

Solve for x : $4/x + 3/x = 7$.

- A) 1
 - B) 7
 - C) $7/4$
 - D) $1/7$
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Question 9 [Medium]

The expression $(x^2 + 7x + 10)/(x^2 - 4)$ can be simplified to which of the following?

- A) $(x + 5)/(x - 2)$
 - B) $(x + 5)/(x + 2)$
 - C) $(x + 2)/(x - 2)$
 - D) $(x - 5)/(x - 2)$
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Question 10 [Medium]

What is the solution to the equation $3/(x + 2) = 1/(x - 4)$?

- A) -7
 - B) 7
 - C) 5
 - D) -5
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Question 11 [Medium]

Simplify: $[(x^2 - 1)/(x + 2)] \cdot [(x + 2)/(x - 1)]$.

- A) $x + 1$
 - B) $x - 1$
 - C) $x^2 - 1$
 - D) $(x + 1)(x + 2)$
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Question 12 [Hard]

Student-Produced Response

For what value of x is the equation $2/(x - 1) + 3/(x + 1) = 10/(x^2 - 1)$ true?

Question 13 [Hard]

Which of the following is equivalent to $[1/a - 1/b] / [1/a + 1/b]$?

- A) $(b - a)/(b + a)$
 - B) $(a - b)/(a + b)$
 - C) $ab/(a + b)$
 - D) $(a + b)/ab$
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Question 14 [Hard]

Student-Produced Response

The function $f(x) = (3x^2 - 12)/(x^2 - 4x + 4)$ can be simplified. What is the value of $f(5)$?

Question 15 [Hard]

Solve: $x/(x - 3) - 2 = 3/(x - 3)$. How many solutions does this equation have?

- A) 0
 - B) 1
 - C) 2
 - D) Infinitely many
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Question 16 [Hard]

Which expression is equivalent to $(4x^2 - 25)/(2x^2 + x - 15)$?

- A) $(2x + 5)/(x + 3)$
 - B) $(2x - 5)/(x - 3)$
 - C) $(2x - 5)/(x + 3)$
 - D) $(2x + 5)/(x - 3)$
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Question 17 [Hard]

The expression $(x^3 + 8)/(x + 2)$ is equivalent to which of the following?

- A) $x^2 + 4$
 - B) $x^2 - 2x + 4$
 - C) $x^2 + 2x + 4$
 - D) $x^2 - 4$
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Question 18 [Hard]

If $3/(a - 3) + B/(a + 3) = (5a + 3)/(a^2 - 9)$ is true for all values of a where the expressions are defined, what is the value of B ?

- A) 1
 - B) 2
 - C) 3
 - D) 5
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Question 19 [Hard]***Student-Produced Response***

The equation $1/(x + 2) + 1/(x - 2) = k/(x^2 - 4)$ has no solution. What is the value of k ?

Question 20 [Medium]

A student completed a problem set. She solved the first half of the problems at a rate of 8 problems per hour and the second half at a rate of 12 problems per hour. If the problem set had p problems, which of the following expresses the total time, in hours, the student spent?

- A) $p/20$
 - B) $5p/48$
 - C) $p/4$
 - D) $20/p$
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